

Real Estate Shocks and Financial Advisor Misconduct

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ABSTRACT

We test whether personal real estate shocks affect professional misconduct by financial advisors. We use a panel of advisors' home addresses and examine within-advisor variation relative to other advisors who work at the same firm and live in the same ZIP code. We find a negative relation between housing returns and misconduct. We show that advisors' housing returns explain misconduct against out-of-state customers, breaking the link between customer and advisor housing shocks. Furthermore, the results are stronger for advisors with lower career risk from committing misconduct, and for advisors with greater borrowing constraints.

THE MAJORITY OF INDIVIDUAL INVESTORS consult an advisor when making financial decisions, and studies show that advisors strongly influence household portfolio choice.^{1,2} However, recent studies such as Dimmock, Gerken, and Graham (2018) and Egan, Matvos, and Seru (2019) show that misconduct by financial advisors is common, and many advisors exploit their clients through

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¹ See Hung et al. (2008) and Foerster et al. (2017).

² For discussions of the relation between trust in financial advisors and household investment, see Burke and Hung (2015), Gennaioli, Shleifer, and Vishny (2015), Giannetti and Wang (2016), and Gurun, Stoffman, and Yonker (2018).

activities such as churning, unauthorized trading, and selling unsuitable investments. This misconduct harms households' financial well-being, through both the direct damages suffered and indirect damages such as reduced investment caused by increased mistrust of financial advisors. For regulators and investors to limit such harm, it is important to understand the determinants of misconduct.

Prior studies show that advisors' type affects misconduct—some advisors have a persistent tendency to commit misconduct (see Egan, Matvos, and Seru (2019)). Other studies show that changes in advisors' professional circumstances affect misconduct (e.g., changes in regulatory oversight, learning from coworkers, and incentive payments).³ A broader literature on corporate fraud examines how work-related incentives affect work-related misconduct. In this paper, we focus instead on *personal* circumstances as a determinant of professional misconduct.⁴ Specifically, we test whether housing price shocks cause financial advisors to commit professional misconduct.

Conceptually, housing prices could have a positive or a negative relation with misconduct. On the one hand, misconduct is a risky activity and so, under decreasing absolute risk aversion, a decrease in wealth should increase sensitivity to risk and reduce misconduct. On the other hand, a key aspect of misconduct is that it produces immediate financial returns, while the costs—if they occur—occur in the future. Thus, as Andreoni (1992) shows, misconduct can function as a form of risky borrowing. In our setting, an advisor who wishes to borrow can compare the costs of conventional borrowing against those of misconduct. Changes in collateral value, such as housing prices, change the costs of conventional borrowing and hence directly affect the relative attractiveness of borrowing via misconduct. Likewise, ethical preferences can produce a directionally similar effect. As Block and Heineke (1975) show, if ethical behavior is a normal good, then fluctuations in wealth will affect an individual's willingness to commit misconduct.⁵

Ultimately, whether changes in housing prices cause misconduct is an empirical question. In this paper, we exploit within-advisor and within-firm-year variation in housing returns to identify whether wealth shocks affect financial advisors' propensity to engage in misconduct.

To measure personal housing wealth shocks and professional misconduct, we use detailed mandatory disclosure filings made by financial advisors, which identify financial misconduct committed by specific advisors as well as their

³ For work on regulatory oversight, see Charoenwong, Kwan, and Umar (2019), for work on learning from coworkers, see Dimmock, Gerken, and Graham (2018), and for work on incentive payments, see Bergstresser, Chalmers, and Tufano (2009), Hackethal, Haliassos, and Jappelli (2012), Mullainathan, Noeth, and Schoar (2012), Chalmers and Reuter (2020), Hoechle et al. (2018), and Egan (2019).

⁴ Bergstresser and Philippon (2006) and Burns and Kedia (2006) examine executive compensation and accounting misstatements.

⁵ Block and Lind (1975) provide an alternative explanation for a relation between wealth shocks and misconduct, arguing that the financial penalties associated with detected misconduct provide little deterrent for an individual close to bankruptcy.

employment histories (see Dimmock, Gerken, and Graham (2018) and Egan, Matvos, and Seru (2020, 2019)). These mandated filings also include advisors' home addresses and dates of residency. We combine the advisor addresses with ZIP-code-level house price indexes created by Zillow to impute housing returns for each advisor. Using the imputed housing returns, we construct each advisor's *cumulative* return since purchase, which Gerardi et al. (2018) argue is a good measure of the exogenous component of loan-to-value (LTV) ratios. In validation tests, we show that cumulative returns predict actual bankruptcy filings and underwater home sales by the advisors in our sample. In addition, back-of-the-envelope calculations show that real estate shocks are large relative to advisor wealth—a one standard deviation change in annual housing returns is approximately equivalent to a 17% change in net wealth. A large literature shows that housing price fluctuations have significant effects on consumption, and that much of this effect occurs through the collateral channel.⁶ Thus, house price returns are economically important, especially for borrowing decisions.

Identifying whether cumulative housing returns affect misconduct is complicated by possible confounding variation. For example, the overall economy could affect both housing returns and misconduct. We rely on multiple fixed effects for identification. Specifically, our baseline specification includes advisor, firm-year, residency-length decile, and ZIP code fixed effects. The advisor fixed effects remove the advisor's overall propensity to commit misconduct, the advisor's "type," as well as individual characteristics such as gender, education, and religious background. The firm-year fixed effects control for firm characteristics such as product offerings and internal oversight (even if these effects are time-varying), as well as for any time-series changes that affect all advisors such as the overall economy. The ZIP code fixed effects remove the effects of characteristics specific to the area, such as demographics, state-level regulation, and local culture (e.g., Parsons, Sulaeman, and Titman (2018)).

We find a strong negative relation between cumulative returns and misconduct: advisors are more likely to commit misconduct after negative housing price shocks and less likely after positive shocks. The results imply that a one standard deviation decrease in cumulative returns is associated with a 13% increase in the likelihood of misconduct (relative to the baseline misconduct rate).

During our 1999 to 2013 sample period, the largest housing market shock occurred around the 2008 financial crisis. Accordingly, as a robustness test we estimate a differences-in-differences model using housing price changes during the financial crisis. The results show that advisors who suffer large housing price declines subsequently increase misconduct. For example, advisors who suffer a price decline of 10% increase misconduct by 23.4% relative to

⁶ For example, see Campbell and Cocco (2007), Gan (2010), Carroll, Otsuka, and Slacalek (2011), Mian and Sufi (2011), Mian, Rao, and Sufi (2013), Bhutta and Keys (2016), Aladangady (2017), DeFusco (2018), and Chen, Michaux, and Roussanov (2020). See Berger et al. (2018) for a theoretical explanation of this empirical relation.

other advisors. In other tests, we show that housing price movement outside of the financial crisis also affects misconduct. As a further robustness test, we estimate panel regressions that use annual rather than cumulative housing returns. Consistent with the baseline specification, we find that annual returns continue to be inversely associated with misconduct.

A potential concern is that local time-varying confounding variation, such as the local economy, could affect both misconduct and housing returns. To address this concern, we exploit the fact that *cumulative* housing returns can vary across advisors living in the same ZIP code at the same time depending on the time of purchase. This variation allows us to include ZIP code-year fixed effects, which remove any local time-varying confounding variation such as shocks to the home prices of the local customer base. In an additional test, we include branch-ZIP-year fixed effects. In this specification, the fixed effects limit the comparison to advisors who live in the same ZIP code during the same year and also work in the same county at the same firm. Even with these more stringent fixed effects, we continue to find that negative cumulative returns are associated with higher misconduct.

We further alleviate concerns related to the commonality of advisor and local customer shocks by focusing on misconduct reported by nonlocal parties. First, we limit the dependent variable to include only instances of misconduct in which the advisor and the customer live in different states. Second, we define misconduct as employment termination or regulatory sanctions. We exclude all advisor-year observations that include a customer-driven complaint to ensure that this alternative dependent variable is distinct from the primary dependent variable. We also employ firm-year fixed effects for employment terminations and state-year fixed effects for regulatory sanctions to control for time-varying characteristics of the complainants. Using these alternative dependent variables, we continue to find a significant negative relation between cumulative returns and misconduct in a setting where the advisor and the complainant are not exposed to the same shock.

We next examine cross-sectional variation in the risk that an advisor is terminated for misconduct. Egan, Matvos, and Seru (2020, 2019) document large variation in the likelihood an advisor is terminated after committing misconduct—some firms are more tolerant of misconduct, and women are punished more severely than men. All else equal, higher career risk reduces the incentive to commit misconduct. Consistent with this intuition, the relation between cumulative housing returns and misconduct is weaker for advisors with higher career risk. We also show that our results are robust to alternative definitions of the dependent variable. Most importantly, we decompose the dependent variable into (i) *active* misconduct such as churning and unauthorized trading, which involves advisors deliberately exploiting clients for financial gain, and (ii) *passive* misconduct such as negligence, which involves advisors harming clients through inattention. We find some evidence of an effect on passive misconduct, but significant evidence of an effect on *active* misconduct, indicating that housing returns affect advisors' deliberate actions.

Our paper is the first to show that changes in employees' personal circumstances affect professional misconduct. The final empirical tests in the paper examine the potential mechanisms that drive this relation. We argue that the evidence is most consistent with mechanisms based on collateral constraints or ethical costs.

Lott (1990) argues that, in the absence of pledgeable collateral, misconduct is the only available borrowing mechanism. Andreoni (1992) and Kotsogiannis and Mateos-Planas (2019) formalize this idea and show that if the rewards from misconduct occur before the potential penalties, collateral values have an inverse relation with misconduct.⁷ In our setting, advisors who wish to borrow can choose between conventional loans, such as home equity extraction, and "borrowing" via misconduct. The availability of home equity is relevant for this choice, as housing equity has a large effect on households' borrowing and consumption. The literature generally finds that households consume out of housing equity throughout the full range of housing returns (i.e., even households with high positive cumulative returns consume out of home equity), but the propensity to consume out of home equity is larger for households with greater financial constraints.⁸ Accordingly, we expect to find a negative relation between housing returns and misconduct over the entire range of returns, with the relation stronger for advisors with greater constraints. This differs from many other possible mechanisms that imply a sharply discontinuous relation with little observable effect for large positive cumulative returns (e.g., debt overhang as in Melzer (2017) and Bernstein (2021) or psychological distress as in Maturana and Nickerson (2020)).

Although we cannot directly observe advisors' borrowing constraints, we create two proxies: advisors with large negative cumulative returns and advisors with high imputed LTV ratios. We create interaction terms between the advisor's cumulative return and the proxies for borrowing constraints. For both interaction terms, the coefficient is negative and significant, indicating that the relation between housing returns and misconduct is stronger for advisors with greater borrowing constraints. Importantly, the relation between housing returns and misconduct remains significant over the entire range of housing returns even with the inclusion of the interaction terms. These findings are therefore consistent with borrowing constraints or ethical preferences, but inconsistent with many alternative mechanisms.

Our paper is related to several recent studies that examine how housing price shocks affect professional risk-taking behavior. Pool et al. (2019) show that mutual fund managers who suffer negative housing shocks subsequently reduce portfolio risk and tracking error. Bernstein, McQuade, and Townsend (2021) show that workers who suffered larger losses on their house values

⁷ Empirical studies such as Jacob, Lefgren, and Moretti (2007), Foley (2011), Munyo and Rossi (2015), and Aneja and Avenancio-Leon (2019) find that liquidity constraints cause an increase in crime. These papers differ from ours in that they study economically motivated street crime by welfare recipients and recent parolees, rather than financial misconduct by professionals.

⁸ See Mian, Rao, and Sufi (2013), Aladangady (2017), and Cloyne et al. (2019).

during the financial crisis subsequently undertook less risky and less innovative projects. In these papers, the professional faces a risky choice where the potential payoffs occur in the future. In our setting, in contrast, the professional can benefit *now* at the risk of later consequences.

Our paper contributes to the recent literature on misconduct by financial advisors. Egan, Matvos, and Seru (2019) study how misconduct affects the labor market for financial advisors and show that misconduct is persistent at the individual level, consistent with advisors' "type" affecting misconduct. Similarly, Clifford, Ellis, and Gerken (2019) find that childhood origin is related to advisor misconduct during adulthood. Several papers show that advisor misconduct varies with work-related circumstances. In particular, Dimmock, Gerken, and Graham (2018) show that peer behavior influences misconduct, Charoenwong, Kwan, and Umar (2019) show that regulatory oversight affects misconduct, and Clifford and Gerken (2021) show that the assignment of property rights to client relationships reduces misconduct that harms clients. In our paper, we show that willingness to engage in misconduct is a pliable characteristic: fluctuations in personal finances affect professional misconduct.⁹ Understanding such causes of financial misconduct is necessary for the design of monitoring and regulatory systems.

Our paper also contributes to the broader literature on financial misconduct. Much of the existing literature focuses on work-related features of the opportunity for misconduct, such as incentive payments, economic conditions, and regulatory oversight. In contrast, our study uses fixed effects to remove the work-related variation in the opportunities for professional misconduct. Instead, we focus on how variation in *personal* circumstances, based on fluctuations in housing prices, affects advisors' willingness to exploit *professional* opportunities for misconduct. To our knowledge, we are the first to show how changes in personal circumstances affect professional misconduct.

Many types of financial misconduct, such as tax evasion, accounting misstatement, Ponzi schemes, and overstating of investment returns, share an intertemporal dimension—offenders improve their current situation at the risk of future penalties. That is, misconduct is both a risky choice and an intertemporal choice, whereby misconduct allows the offender to borrow from the future. The intertemporal aspect of misconduct is largely ignored, however, in the literature.¹⁰ Empirically examining misconduct as risky borrowing is challenging because factors that affect the intertemporal trade-off will also often affect the direct cost-benefit trade-off. Our setting has two unique features that allow us to overcome these challenges. First, studying how shocks to personal collateral affect professional misconduct breaks the direct links between the shock and the opportunities for misconduct. Second, the financial advisor data allow for individual-level analysis within firms, which allows

⁹ Kleiner, Stoffman, and Yonker (2021) show a result in the opposite direction—that professional interactions with coworkers can affect personal bankruptcy decisions.

¹⁰ Andreoni (1992) and Kotsogiannis and Mateos-Planas (2019) provide theoretical models of tax evasion as borrowing, but empirical evidence is lacking.

us to remove many possible sources of confounding variation, and thereby provide evidence on the misconduct-as-risky-borrowing channel.

Our results further relate to the literature on the economics of crime and misconduct, particularly the question of whether there are wealth effects in unethical behavior. Many models of crime and misconduct assume that people prefer honesty to dishonesty, and incorporate a utility penalty from committing misconduct. For example, Block and Heineke (1975) model the case of street crime, Gordon (1989) models the case of tax evasion, and McGuire and Pauly (1991) model medical misconduct. Across these diverse settings, the utility penalty implies, under reasonable assumptions, that honesty is a normal good, and that an individual's willingness to commit misconduct is decreasing in wealth. The results in our paper are consistent with honesty as a normal good, which, to our knowledge, has not previously been shown for white-collar professionals.

We also document another important externality of housing price shocks. Aside from the direct wealth effects of housing price shocks, several papers document less obvious effects. For instance, Campbell, Giglio, and Pathak (2011) show that foreclosures have spillover effects that reduce the value of neighboring houses, and Mian, Rao, and Sufi (2013) highlight the decline in consumption following the housing crisis. We identify another externality—investors suffer increased misconduct as a result of their financial advisors' real estate price shocks.

This paper is organized as follows. Section I outlines the data used in the study. Section II discusses the identification strategy. Section III reports our main results. Section IV addresses concerns about commonality of shocks between advisors and their clients. Section V provides various robustness tests. Section VI documents cross-sectional variation. Finally, Section VII concludes.

I. Data and Sample Construction

Our financial advisor data come from a panel of mandatory disclosures made in Form U4 filings. All registered representatives¹¹ in the United States are required to file and update these forms following any material changes. The Financial Industry Regulatory Authority (FINRA) assigns each advisor a unique individual permanent identifier that allows us to track advisors even if they switch employers. We obtain this panel through a combination of Freedom of Information Act (FOIA) requests filed with state regulators, third-party data from Meridian IQ, and the FINRA BrokerCheck website.¹² See Dimmock, Gerken, and Graham (2018) and Egan, Matvos, and Seru (2020, 2019) for further discussion of the data and industry.

¹¹ "Registered representative" is the term FINRA uses for these individuals. Following the literature, we use the term "financial advisor." These individuals are also commonly referred to as brokers or financial planners.

¹² The BrokerCheck website allows investors to access a subset of the data reported in the Form U4 filings. BrokerCheck does not report the advisors' home addresses. See <https://brokercheck.finra.org/>.

The financial advisor data include residential addresses for the 1999 to 2013 period, as well as all misconduct disclosures made over the 1999 to 2017 period. The sample includes 428,108 advisors with complete residential histories. We combine the advisor data with ZIP-level housing price data from Zillow, a real estate data company.¹³ However, because Zillow does not provide the necessary data for all ZIP codes,¹⁴ our final sample contains 329,418 advisors.

A. Financial Advisor Data

The Form U4 data include each advisor's employment history, licenses, qualifications, and any mandated disclosures. Advisors are required to disclose certain information about customer complaints, regulatory actions, civil and criminal legal cases, terminations, and bankruptcies. Table I summarizes this disclosure information. Columns (1), (2), and (3) report the number of advisors with a disclosure, the percentage who make a disclosure at any time during the sample, and the percent of advisor-years with a disclosure, respectively.

Following Dimmock, Gerken, and Graham (2018), our primary measure of misconduct is validated customer complaints. Customer complaints are formal complaints whereby a customer demands compensation for damages caused by an advisor's alleged misconduct. Although some customer complaints allege negligence, most complaints allege improper behavior taken to increase the advisor's compensation. Financial advisors are compensated primarily based on the revenues they generate (see Hung et al. (2008)), which creates incentives for advisors to generate trading commissions or sell products with high distribution fees. Theoretical models show that these incentives encourage misconduct,¹⁵ and empirical tests find support for the models' predictions.¹⁶

After a customer files a complaint, it can be resolved through arbitration, settlement, or withdrawal. Following Dimmock, Gerken, and Graham (2018), we create the variable *Misconduct* based on customer disputes for which either the arbitration panel rules in the customer's favor or the dispute is settled for at least a certain minimum amount.¹⁷ This variable therefore includes only those customer complaints that are validated through an arbitration decision or a sizable monetary payment (i.e., we do not include customer complaints that are withdrawn or unresolved as of the end of our sample). Table I shows that 0.63% of advisor-years and 4.66% of advisors in our final sample report at least one *Misconduct* event.

¹³ For further information about the Zillow data, see <https://www.zillow.com/research/data/>.

¹⁴ Since the Zillow data set is also limited to the United States, we are unable to include foreign addresses.

¹⁵ See Inderst and Ottaviani (2009, 2012) and Stoughton, Wu, and Zechner (2011).

¹⁶ See Bergstresser, Chalmers, and Tufano (2009), Hackethal, Inderst, and Meyer (2011), Hackethal, Haliassos, and Jappelli (2012), Mullainathan, Noeth, and Schoar (2012), Chalmers and Reuter (2020), and Hoechle et al. (2018).

¹⁷ Financial advisors must report settlements of \$10,000 or more before May 19, 2009 and of \$15,000 or more thereafter. Settlements less than these thresholds need not be disclosed, as they are potentially "nuisance" settlements rather than valid complaints.

Table I
Advisors and Disclosures

This table summarizes misconduct and financial disclosures that financial advisors are required to make. The first column reports the number of advisors with the associated misconduct measure. The second and third columns report the percent of offending advisors and advisor-years, respectively. *Misconduct* is defined following Dimmock, Gerken, and Graham (2018) as customer disputes that receive an award in arbitration or are settled for at least a certain dollar value (\$10,000 or \$15,000 depending upon the time period). *Out-of-State Misconduct* is defined using customer disputes from clients who reside in a different state than the advisor. *Regulatory* is a disclosure of finalized regulatory sanctions from entities such as the SEC, FINRA, or state regulators. *Employment Separation After Allegations* indicates whether the advisor was terminated or permitted to resign following allegations of misconduct. We define *Active Misconduct* as customer disputes related to misrepresentation, unauthorized activity, fees/commissions, churning/excessive trading, and fraud. We define *Passive Misconduct* as customer disputes related to negligence and omission of key facts. *Mutual Fund Misconduct* indicates whether the misconduct was related to mutual funds. *≥100k Misconduct* indicates whether alleged damages exceed \$100,000. *EMS Misconduct* is defined using a broader set of disclosures as in Egan, Matvos, and Seru (2019) and includes *Misconduct*, *Regulatory*, and *Employment Separation After Allegations*, as well as certain civil law and criminal disclosures. *Bankruptcy* reports disclosures of bankruptcies by financial advisors. *Underwater Sale* identifies disclosures that pertain specifically to an underwater sale of a property.

	Advisors	Advisors (%)	Advisor-Years (%)
Misconduct	14,691	4.66	0.63
Out-of-State Misconduct	4,016	1.28	0.16
Regulatory	2,535	0.80	0.10
Employment Separation After Allegations	3,420	1.09	0.12
Active Misconduct	7,827	2.48	0.32
Passive Misconduct	2,788	0.89	0.11
Mutual Fund Misconduct	8,645	2.74	0.35
≥100k Misconduct	5,194	1.65	0.21
EMS Misconduct	19,232	6.11	0.85
Bankruptcy	4,773	1.55	0.31
Underwater Sale	1,643	0.53	0.11

Table I also summarizes additional disclosures that we employ in robustness tests. In particular, *Out-of-State Misconduct* includes only those cases of *Misconduct* in which the customer and the advisor live in different states, *Regulatory* reports finalized regulatory sanctions from entities such as the Securities and Exchange Commission (SEC), FINRA, or state regulators, and *Employment Separation After Allegations* indicates whether the advisor is terminated or permitted to resign following allegations of misconduct.¹⁸

In robustness tests, we follow Egan, Matvos, and Seru (2019) and combine *Misconduct* with *Regulatory*, *Employment Separation After Allegations*, and

¹⁸ An *Employment Separation After Allegations* occurs if the advisor is discharged, voluntarily resigns, or is permitted to resign after allegations accusing the advisor of: “(1) violating investment-related statutes, regulations, rules, or industry standards of conduct; (2) fraud or wrongful taking of property; or (3) failure to supervise in connection with investment-related statutes, regulations, rules, or industry standards of conduct.” For more details, see <https://www.finra.org/sites/default/files/AppSupportDoc/p015111.pdf>.

certain civil law and criminal disclosures¹⁹ to create *EMS Misconduct*. Table I shows that 6.11% of advisors in our final sample report at least one event under the broader Egan, Matvos, and Seru (2019) definition.

Table I additionally shows that 1.55% of advisors report a bankruptcy and 0.53% of advisors report a compromise with creditors in relation to selling a home for less than the outstanding mortgage amount (i.e., an underwater sale). We use these variables to check that housing prices relate to financial well-being and liquidity constraints.

Panel A of Table II reports additional summary statistics related to advisor misconduct. For advisors with a settled or awarded customer complaint, the average (median) alleged damages are \$659,666 (\$19,000) and the average (median) settlement amount is \$239,485 (\$15,754). For advisors with regulatory sanctions, the average (median) penalty is \$13,125 (\$5,000). Note that for approximately 40% of the cases we are unable to observe settlement, award, or penalty amounts. In addition to the misconduct information, Table II summarizes additional information from the Form U4 filings. The median advisor has 10 years of industry experience, and in all regressions we control for the logarithm of this variable.

B. Financial Advisors' Homes and Real Estate Price Shocks

Form U4 requires each advisor to report the address and dates of occupancy for each residence throughout the sample period.²⁰ For example, an advisor who has resided in the same house since July of 1960 would report a move-in date of July 1960 even though the sample period does not begin until 1999. We do not observe whether the advisor rents or owns the home. However, robustness tests reported in Internet Appendix Table IA.I that use proxies for the likelihood of ownership suggest that renting is unlikely to materially affect our results.^{21,22}

We filed FOIA requests with all state regulators, but numerous states did not supply home addresses. As a result, we have home addresses only for advisors

¹⁹ The civil law disclosures report court-issued injunctions regarding investment-related activity, findings of a violation of any investment-related statute or regulation, and actions brought by a state or foreign financial regulatory authority that are dismissed by a court pursuant to a settlement agreement. The criminal disclosures report any felony convictions or charges, as well as certain misdemeanors such as bribery, perjury, fraud, or wrongful taking of property.

²⁰ Section 11 of the Uniform Application for Securities Industry Registration (Form U4) asks advisors to "provide their residential addresses for the past five (5) years. Leave no gaps greater than three (3) months between addresses. Begin by entering your current residential address. Enter 'Present' as the end date for your current address. Post Office boxes are not acceptable. Report changes as they occur."

²¹ If measurement error from assigning housing returns to renters is pure measurement error, this would result in attenuation bias and our results would understate the true effect. If, however, renting is disproportionately associated with areas in which there is a negative correlation between housing prices and misconduct, this could bias our results upward (although it is not obvious why this correlation pattern would occur).

²² The Internet Appendix may be found in the online version of this article.

Table II
Summary Statistics

This table reports summary statistics. Panel A reports statistics regarding advisor misconduct. Panel B reports statistics regarding advisor residency and experience for the 2,860,572 advisor-year observations. Panel C summarizes the distribution of house price changes.

Panel A: Advisor Misconduct							
	1%	25%	Median	Mean	75%	99%	
Alleged Damages from Customer Disputes (\$)	0	5,000	19,000	659,666	100,000	3,500,000	
Settle Amount for Customer Disputes (\$)	0	2,870	15,754	239,485	59,347	3,100,000	
Regulatory Damages (\$)	100	1,000	5,000	13,125	10,000	200,000	
Panel B: Advisor-Year Variables							
	1%	25%	Median	Mean	75%	99%	
Industry Experience (years)	1	5	10	11.8	17	37	
House Purchase Price (\$)	65,800	153,100	229,700	320,539.0	376,300	1,502,300	
Zillow ZIP Price (\$)	73,800	176,000	268,900	373,678.6	445,700	1,714,400	
Years at Residence	1	3	5	5.8	8	22	
Panel C: Distribution of House Price Changes							
	1%	5%	10%	Median	90%	95%	99%
Annual Dollar Change	-91,300	-42,600	-26,000	3,700	49,000	76,400	164,100
Annual Percent Change	-18.1	-10.7	-7.6	1.7	13.7	17.8	26.7
Cumulative Dollar Change	-212,200	-97,900	-54,000	23,500	198,900	303,800	618,800
Cumulative Percent Change	-37.9	-20.9	-13.1	9.7	64.1	92.1	150.0

who register in the District of Columbia, Florida, Georgia, Hawaii, Indiana, Iowa, New Jersey, Oregon, Rhode Island, Tennessee, Texas, Washington, or West Virginia. Because advisors must register with each state in which they do business, and many advisors register in all states, we have home addresses in all 50 states (e.g., we have home addresses for 69,167 advisors who live in New York but also register in a state that provided data). Thus, although our sample of home addresses is not comprehensive, it covers all major real estate markets. Importantly, the selection mechanism is the state regulator's interpretation of The Privacy Act of 1974 as it relates to our FOIA request. It is not obvious that this selection mechanism would be systematically related to the correlation between local housing shocks and misconduct by financial advisors, and thus we do not see reasonable concerns related to selection bias.

We match the advisor residences to the Zillow ZIP code house price indexes. The combined advisor-residence-Zillow price data set spans 13,679 ZIP codes, which collectively contain 74.7% of the U.S. population. Using the Zillow ZIP code indexes, we impute a purchase price and annual price changes for each advisor-residence combination.²³ This approach is similar to that of Bernstein, McQuade, and Townsend (2021), who also impute individual house price shocks using ZIP-level price indexes.²⁴ In Internet Appendix Table IA.II, we use the House Price Index produced by the Federal Housing Finance Agency²⁵ as an alternative measure of real estate prices and find similar results. For advisors who report multiple residences at the same time (e.g., vacation homes), we use purchase-price-weighted average returns. In Internet Appendix Table IA.III, we show that our results are robust to using only the highest-value residence or excluding all advisor-years with multiple residences.

Panel B of Table II shows that the average house in the sample has an imputed purchase price of \$320,539 and a current price of \$373,679. On average, the advisors have lived in their current house for 5.8 years. Panel C reports the distribution of annual house price changes. The median annual price change is \$3,700, which is 1.72% of the beginning-of-year value. As the percentiles show, there is considerable variation in returns.

We use the imputed purchase price and annual returns to calculate the *Cumulative Return* for each advisor-year. Importantly, this varies across advisors, even within a fixed ZIP code, depending on when the advisors purchased their house. Panel C of Table II shows that the median *Cumulative Return* is 9.7% and the variation is large.

We use cumulative returns as the main independent variable for several reasons. First, Gerardi et al. (2018) use cumulative returns as an instrument for

²³ Pool et al. (2019) use house-level imputed values from Zillow, instead of ZIP-level imputation. This is feasible in their study as they have fewer than 1,000 individuals. This approach is not feasible in our context given our large sample size. We obtain residence-specific Zillow valuation estimates for a subsample of advisors and find that the average time-series correlation between the residence-specific price changes and the ZIP code price changes is 0.803.

²⁴ Bernstein, McQuade, and Townsend (2021) use the indexes developed by Bogin, Doerner, and Larson (2019) but report that robustness results using the Zillow indexes are similar.

²⁵ See <https://www.fhfa.gov/DataTools/Downloads/pages/house-price-index.aspx>.

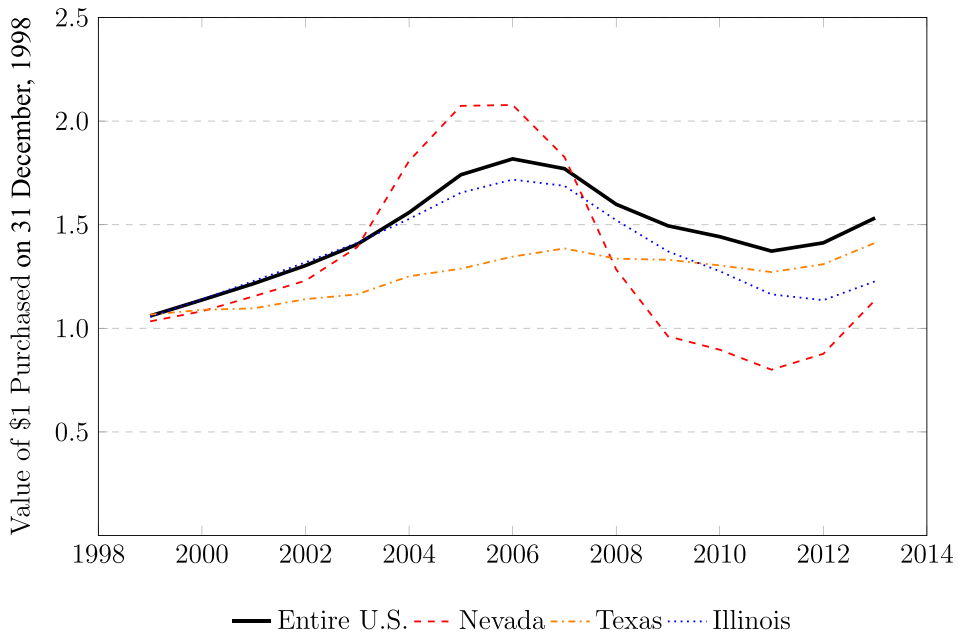


Figure 1. Real estate returns over sample period. This figure plots the cumulative return as measured by the Zillow national- and state-level house price indexes over the sample period. We plot this measure separately for the entire United States (black) and select states. (Color figure can be viewed at wileyonlinelibrary.com)

LTV ratios and show that the two variables are highly correlated. They further argue that using cumulative returns avoids endogeneity biases from households' choices related to down payment size, mortgage type, prepayments, etc. Second, the salience of the purchase price makes it a natural basis for comparison with current prices. Third, because of variation in the timing of purchases, cumulative returns vary across advisors within a ZIP code in any given year, which allows for the inclusion of more stringent fixed effects in some tests.

Figure 1 summarizes the time series and regional variation in real estate prices. The figure plots the *Cumulative Return* during our sample period for the United States and for selected states using Zillow Home Value indexes. As can be seen, there is a clear time-series pattern with rising prices in the early sample period and a decline around the housing market crash of 2007 to 2009, although there is variation across states in the timing of the crash. For example, prices in Nevada began to decline before other states. There is also considerable cross-sectional variation. For example, home prices in Nevada vary much more than in Texas.

We also find large cross-sectional variation within states and even within metropolitan statistical areas (MSAs), as documented by Edlund, Machado, and Sviatschi (2016) and Bogin, Doerner, and Larson (2019). For example, Figure 2 displays ZIP-level housing price returns for one particular metropolitan

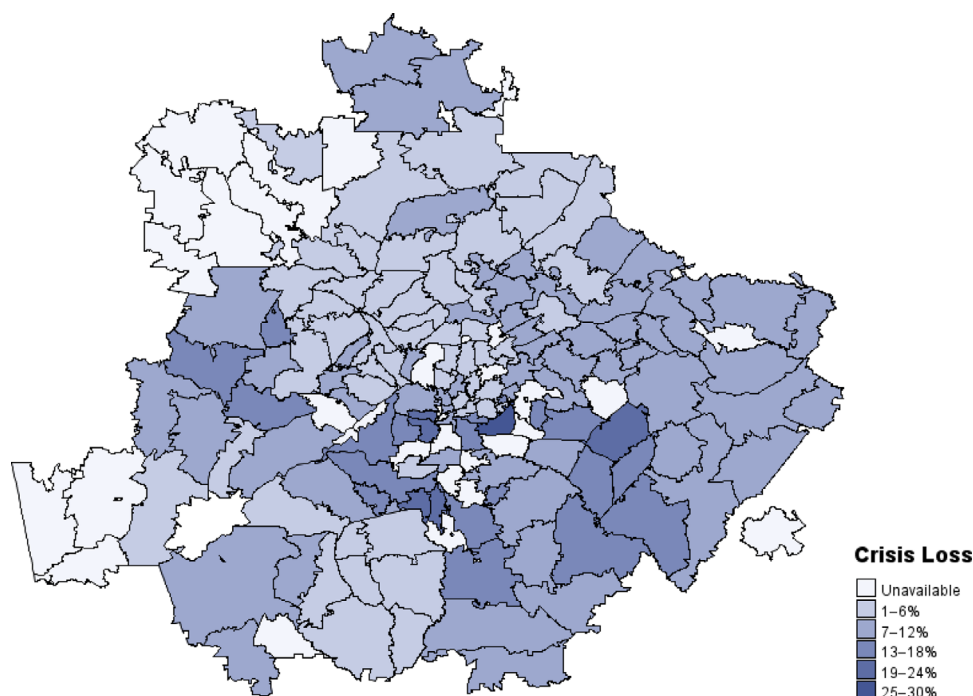


Figure 2. Crisis price changes in metro Atlanta. This figure displays price changes by ZIP code in 2008 for the Atlanta, GA metropolitan statistical area. ZIP codes are color coded by level of 2008 loss, where darker shades indicate more severe losses. (Color figure can be viewed at wileyonlinelibrary.com)

area (Atlanta) in 2008: the hardest-hit ZIP codes lost as much as 27%, while other nearby ZIP codes had negligible losses.

II. Identification Strategy

Identifying whether housing shocks cause financial advisors to commit misconduct is complicated because various unobservable factors affect both housing prices and misconduct. To remove such potentially confounding variation, we use multiple fixed effects.

Our baseline specification is

$$\begin{aligned} Misconduct_{i,t} = & \beta \cdot Cumulative\ Return_{i,t} + \gamma \cdot \log(Exp.)_{i,t} + \delta_i \\ & + \delta_{rl,t} + \delta_{f,t} + \delta_z + \epsilon_{i,t}, \end{aligned} \quad (1)$$

where $Misconduct_{i,t}$ is an indicator variable denoting advisors with a validated customer complaint during the current year. In all regressions with a binary dependent variable, we set the variable to zero or 100 to enhance readability of the tables. $Cumulative\ Return_{i,t}$ is the cumulative price change since

purchase for advisor i and $\log(\text{Exp.})_{i,t}$ is the logarithm of years of industry experience. The specification also includes advisor, residency-length decile $\times$ year, firm-year, and ZIP code fixed effects.

First, the financial advisor fixed effects, δ_i , remove all time-invariant characteristics of the advisor, including the overall propensity to commit misconduct, and reduces the effect of advisor characteristics that are largely fixed over the sample period, such as education and religious background. These fixed effects also remove the time-invariant part of the advisor's business activities, such as customer characteristics and products sold, as well as time-invariant real estate preferences, such as the advisor's propensity to use nonstandard mortgage loans or extract home equity via loans. Including δ_i effectively means that our key independent variable is the within-advisor change in *Cumulative Return* and not its level.

Second, the residency length $\times$ year fixed effects, $\delta_{rl,t}$, remove variation across advisors based on how long they have lived in their current residence. The average LTV ratio varies with the length of residency and recent price paths, and differences in leverage may affect financial advisors' actions. By including these fixed effects, we remove the average effect that residency length $\times$ year has on LTV ratios. In general, *Cumulative Return* is increasing in residency length, but this relation (and changes in its time-series average) is removed by the inclusion of these fixed effects. Furthermore, longer residency may be associated with deeper community ties, increasing the reputational cost of engaging in misconduct.

Third, the firm-year fixed effects, $\delta_{f,t}$, remove the time-invariant characteristics of the firm that employs the advisor, as well as time-varying firm characteristics such as changes in the firm's product offerings or monitoring procedures. Removing firm effects is important as Egan, Matvos, and Seru (2019) show that certain firms specialize in committing misconduct. The firm-year fixed effects also subsume year effects, removing the common time-series variation in housing returns and misconduct (e.g., any common time-series relation between asset prices and complaint rates).

Fourth, the ZIP code fixed effects, δ_z , remove the time-invariant characteristics of the area in which the advisor lives, including the decision by the advisor to live within the ZIP code. This is important as Parsons, Sulaeman, and Titman (2018) provide evidence of large geographic variation in misconduct rates. These fixed effects also remove stable demographic and economic characteristics of the neighborhood.

The key independent variable of interest is the advisor's *Cumulative Return* since purchase, expressed as a percentage of the initial purchase price. Our use of cumulative returns is similar to that of Gerardi et al. (2018), who use cumulative returns as an instrumental variable to estimate the exogenous component of LTV ratios. The *Cumulative Return* isolates the component of real estate wealth (and household mortgage leverage) that is driven by local house price fluctuations, which are not directly under the control of the advisor. For example, *Cumulative Return* is unaffected by the advisor's choice of mortgage type or the use of home equity loans. The only choice components of

the advisor's *Cumulative Return* are determined by the location and timing of purchase; the effect of location choice is removed by the ZIP code fixed effects and the effect of the timing choice of purchase is removed by the residency length $\times$ year fixed effects.

III. Main Results

Although our main specification is based on individual fixed effects and *Cumulative Returns*, we begin our empirical results with relatively simple specifications for ease of interpretation. We begin with a differences-in-differences specification focusing on the financial crisis. We then transition to a panel setting and discuss an annual return shock specification before moving on to the main specification used in the remainder of the paper. All three approaches produce similar results and inferences.

A. Differences-in-Differences

During our sample period, the largest housing price shocks occurred during the 2008 financial crisis. We begin by focusing on this period using a differences-in-differences approach similar to that in Bernstein, McQuade, and Townsend (2021) and Pool et al. (2019). Column (1) of Table III reports the results. In this specification, there is one observation per advisor. The dependent variable is the change in misconduct, defined as the number of instances of misconduct over the three-year period 2008 to 2010 less the number of instances over the period 2005 to 2007.²⁶ The key independent variable is *Crisis Return*, which is the percentage price change in the advisor's house in 2008.²⁷ The control variables are also measured as of 2008 (e.g., the advisor's 2008 employer is used for the firm fixed effects). The regression includes a control for the logarithm of years of industry experience. The specification includes firm and residency length fixed effects. The standard errors are clustered by ZIP code.

The results show a significant negative relation between housing returns and changes in misconduct; advisors with relatively bad housing returns commit relatively more misconduct. The economic magnitude implied by the results is large: the coefficient estimate implies that an advisor whose house price declined by 10% during the crisis (nearly a quarter of advisors suffer such as loss) increases misconduct by 23.4% relative to the baseline frequency of misconduct. These magnitudes are economically significant and comparable to other determinants of the malleability of financial advisor behavior (e.g.,

²⁶ In column (1) of Table III, we include all advisors from 2008 regardless of whether they survive until 2010. In Internet Appendix Table IA.IV, we estimate a similar version in which we require the advisor to be in the sample for all three postcrisis years. The results are similar to those presented in the paper.

²⁷ In Internet Appendix Table IA.V, we replace the continuous price change measure with indicator variables representing losses of 10% and 20%. The results are similar to those reported in the paper.

Table III
House Price Return and Misconduct

This table reports results from regressions of misconduct on an advisor's housing returns. In column (1), the unit of observation is the advisor and the dependent variable is the change in misconduct between the pre- and postperiods, that is, the number of misconduct events in 2008, 2009, and 2010 minus the number of misconduct events for the same advisor in 2005, 2006, and 2007. In columns (2) and (3), the unit of observation is the advisor-year and the dependent variable is an indicator for advisor misconduct (multiplied by 100). *Crisis Return* is the percentage price change in the advisor's house during the financial crisis (the house return in 2008). *Annual Return* is the year-over-year percentage price change in the advisor's house. *Cumulative Return* is the aggregate cumulative price change divided by the purchase price. $\log(\text{Industry Experience})$ is the logarithm of the number of years an advisor has worked in the industry. Indicators for fixed effects are reported beneath the estimates. Standard errors, clustered by advisor and ZIP code, are reported in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% level, respectively.

	Diff-in-Diff (1)	Annual (2)	Cumulative (3)
Crisis Return	-0.0509*** (0.0121)		
Annual Return		-0.1896** (0.0773)	
Cumulative Return			-0.2264*** (0.0334)
$\log(\text{Industry Experience})$	0.0050*** (0.0011)	0.3784*** (0.0069)	0.4869*** (0.0225)
Advisor FE	No	No	Yes
Residency Length $\times$ Year FE	Yes	Yes	Yes
Firm $\times$ Year FE	No	Yes	Yes
Firm FE	Yes	Subsumed	Subsumed
ZIP FE	No	Yes	Yes
R^2	0.025	0.025	0.146
Observations	248,432	3,910,975	2,860,572

slightly larger than going from an unethical [25th percentile] to an ethical [75th percentile] city (Parsons, Sulaeman, and Titman (2018)) and about two-thirds the impact of being exposed to a bad peer (Dimmock, Gerken, and Graham (2018)).

One concern with using a differences-in-differences approach is the possibility that the treated group (advisors who experience a shock during the crisis) are following a different trend than those in the control (advisors who do not experience a shock during the crisis). To address this concern, we estimate a series of annual regressions that predict misconduct in year t using an indicator variable for advisors who suffer a -10% or worse housing return in 2008. We plot these coefficients in Figure 3. The results show that 2008 housing returns do not significantly predict misconduct prior to 2008, but do predict misconduct in 2008 and for several years thereafter. This finding provides comfort that the two groups experienced similar trends during the precrisis period.

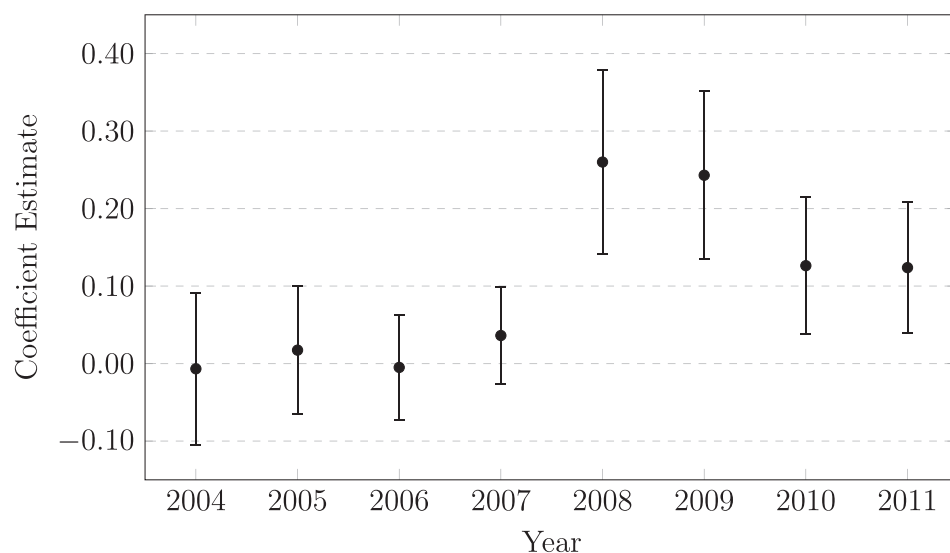


Figure 3. Predicting misconduct using crisis returns. This figure plots estimates and confidence intervals from regressions that use returns in 2008 to predict misconduct in year t , where $CrisisPriceDrop(10\%)_{i,2008}$ is an indicator variable equal to one for advisors who lost at least 10% of the value of their home in 2008:

$$Misconduct_{i,t} = \beta \cdot CrisisPriceDrop(10\%)_{i,2008} + \log(IndustryExperience)_{i,t} + \delta_{f,t} + \delta_{rl} + \epsilon_{i,t}.$$

B. Annual Returns

Next, we generalize from the crisis to examine the full panel of data in which the unit of observation is the advisor-year. The dependent variable is an indicator variable denoting advisors who commit misconduct in the current year. The key independent variable is *Annual Return*, the annual percentage return on the advisor's house. This specification shows the effect of changes in housing wealth on the level of misconduct. The specification includes residency length $\times$ year fixed effects, firm-year fixed effects, and ZIP code fixed effects.²⁸ The standard errors are clustered by advisor and ZIP code. Consistent with the differences-in-differences results, the results in column (2) of Table III show a significant negative relation between *Annual Return* and misconduct.

C. Cumulative Returns

Column (3) of Table III presents the baseline specification used throughout this paper. In this specification, the unit of observation is the advisor-year. The dependent variable is an indicator variable denoting advisors who commit

²⁸ Internet Appendix Table IA.VI shows that the results are similar if the specification includes advisor fixed effects.

misconduct in the current year. The key independent variable is *Cumulative Return*, which is the cumulative percentage return on the advisor's house since purchase. Similar to column (2), this specification includes residency length fixed effects, firm-year fixed effects, and ZIP code fixed effects. Column (3) also includes advisor fixed effects. With the inclusion of advisor fixed effects, the specification effectively shows the relation between changes in an advisor's *Cumulative Return* and changes in the advisor's misconduct. This specification is therefore similar to that in column (2), except that the dependent variable is effectively the difference in misconduct rather than the level.

The results show that there is a negative and significant relation between *Cumulative Return* and misconduct. In particular, the results imply that a one standard deviation decrease (increase) in cumulative returns (35.8 percentage point change) results in a 7.5 basis point increase (decrease) in the likelihood of misconduct, which is a 13% change relative to the baseline misconduct rate. Importantly, all three approaches reported in Table III produce consistent results.

Several tables in the [Internet Appendix](#) report results of numerous robustness tests of the baseline specification in column (3) of Table III. [Internet Appendix Table IA.VII](#) shows that the results are robust to using an indicator variable for misconduct occurring in the subsequent year or during a three-year window. [Internet Appendix Table IA.VI](#) shows that the results hold if we do not include advisor fixed effects and instead include additional control variables such as advisor age. [Internet Appendix Table IA.VIII](#) includes controls for time-varying advisor and location characteristics. [Internet Appendix Table IA.IX](#) excludes the financial crisis period from our sample, as many advisors suffered large negative returns during this period. [Internet Appendix Table IA.X](#) aggregates returns and misconduct to the branch level before estimating results. In all cases, we find similar results as in the baseline specification.

D. A Placebo Test

As a robustness test, we employ a bootstrap-placebo procedure. For each repetition of this procedure, we randomly match each ZIP code with another ZIP code from a different state. We then assign each advisor-year the cumulative housing return from the matched ZIP code-year based on the advisor's year of purchase. The match between ZIP codes is fixed for all years within a single repetition of the bootstrap procedure. For example, suppose that in 2006 an advisor named Bob lived in a house in ZIP code 48823 that he purchased in 2000. Further suppose that, for this repetition of the bootstrap, ZIP code 48823 was randomly assigned returns from ZIP code 78722. In 2006, Bob would be assigned the six-year cumulative housing return from ZIP code 78722. If Bob continued to live in the same house, in 2007 he would be assigned the seven-year cumulative housing return from ZIP code 78722. Other advisors in ZIP code 48823 would also be assigned returns from ZIP code 78722, but their cumulative housing returns would vary depending on their year of purchase.

Using these pseudo-cumulative housing returns, we estimate the specification reported in column (3) of Table III. We repeat this procedure 10,000 times.

This procedure assigns random cumulative returns for each advisor-year observation, but, crucially, preserves all other time-series and cross-sectional relations in the panel. The time-series relations are preserved because the advisor's pseudo-cumulative return is drawn from the same randomly assigned ZIP code for each year. The cross-sectional relations between advisors in a single ZIP code are preserved because their pseudo-cumulative returns are all drawn from the same randomly assigned ZIP code. Furthermore, any effects due purely to the length of residency are preserved. For example, suppose that advisors with greater length of residency generally: (i) have higher cumulative returns and (ii) are less likely to change misconduct following house price shocks because of deeper community ties. This relation would be preserved in the placebo test because the pseudo-cumulative return is a function of length of residency.

Figure 4 plots a histogram of the coefficient estimates on the *Pseudo-Cumulative Return* variable. The actual coefficient estimate of -0.2264 from Table III, column (3) lies over five standard deviations below the mean of the bootstrapped coefficients (-0.007), and none of the 10,000 placebo estimates are below the actual coefficient estimate. Thus, the placebo results suggest that it is the advisor's *actual* returns that affect misconduct.

IV. Addressing Concerns about Commonality in Customer and Advisor Shocks

As in most studies of fraud and misconduct, we observe *detected*, not *actual*, misconduct. This raises the possibility of bias if variation in the detection rate is correlated with the independent variable of interest. In this section, we present a number of tests that address the concern that customers' propensity to file complaints might vary with their real estate returns.

A. ZIP Code-Year Fixed Effects

Column (1) of Table IV, Panel A is similar to the baseline specification but replaces ZIP code fixed effects with ZIP code-year fixed effects. This fixed effects approach exploits the fact that cumulative housing returns vary across advisors in the same ZIP code during the same year based on when the advisors purchased their homes combined with the price path of housing in that ZIP code. The ZIP code-year fixed effects remove variation that is common to all advisors in the same ZIP code during the year, such as local housing price shocks during the year, the economic and demographic characteristics of the local customer base, and any other local commonalities including the propensity of local customers to file complaints. In this specification, the variation in the key variables is limited to the time-series change for an individual *relative* to the average time-series change of other advisors who live in the same ZIP code in that year. For *Cumulative Return*, the variation comes

Table IV
Alternative Fixed Effects, Out-of-State Customers, and Noncustomer Misconduct

This table reports estimates from regressions of measures of misconduct on an advisor's cumulative house price return since purchase. The unit of observation is the advisor-year. Panel A reports estimates with varying levels of fixed effects while Panel B reports estimates using modified versions of the dependent variable. In Panel A, the dependent variable is an indicator for advisor misconduct. In column (1) of Panel B, the dependent variable is an indicator for advisor misconduct reported by clients who live in a different state than the advisor. In column (2) of Panel B, the dependent variable is an indicator for advisor misconduct reported by clients who live in a state that does not border the advisor's state. In columns (3) and (4) of Panel B, the dependent variable is an indicator for regulator- or firm-reported incidences of misconduct. In column (4) of Panel B, we exclude sample observations if a client also reports an incident of misconduct in the same year. *Cumulative Return* is the aggregate cumulative price change divided by the purchase price. $\log(\text{Industry Experience})$ is the logarithm of the number of years an advisor has worked in the industry. In each column, we multiply the dependent variable by 100. Indicators for fixed effects are reported beneath the estimates. Standard errors, clustered by advisor and ZIP code, are reported in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% level, respectively.

Panel A: Additional Fixed Effects				
	(1)	(2)	(3)	(4)
Cumulative Return	-0.1560*** (0.0430)	-0.1388*** (0.0431)	-0.1812*** (0.0618)	-0.1928** (0.0861)
$\log(\text{Industry Experience})$	0.4992*** (0.0236)	0.5077*** (0.0290)	0.5031*** (0.0353)	0.4546*** (0.0525)
Advisor FE	Yes	Yes	Yes	Yes
Residency Length $\times$ Year FE	Yes	Yes	Yes	Yes
ZIP FE	Subsumed	Yes	No	Subsumed
Firm $\times$ Year FE	Yes	Subsumed	Subsumed	Subsumed
Firm $\times$ Year $\times$ City FE	No	No	Yes	Subsumed
ZIP $\times$ Year FE	Yes	No	No	Subsumed
Branch $\times$ Year FE	No	Yes	No	Subsumed
Branch $\times$ Year $\times$ ZIP FE	No	No	No	Yes
R^2	0.188	0.241	0.358	0.491
Observations	2,833,467	2,312,551	1,975,599	1,076,138
Panel B: Nonlocal Misconduct				
	Out-of-State Customer	Distant Customer	Regulator or Firm Actions	
	(1)	(2)	(3)	(4)
Cumulative Return	-0.0628*** (0.0160)	-0.0446*** (0.0135)	-0.0593*** (0.0208)	-0.0520*** (0.0200)
$\log(\text{Industry Experience})$	0.1065*** (0.0103)	0.0749*** (0.0087)	0.0569*** (0.0141)	0.0521*** (0.0137)
Advisor FE	Yes	Yes	Yes	Yes
Residency Length $\times$ Year FE	Yes	Yes	Yes	Yes
Firm $\times$ Year FE	Yes	Yes	Yes	Yes
ZIP FE	Yes	Yes	Yes	Yes
State $\times$ Year FE	No	No	Yes	Yes
R^2	0.140	0.140	0.176	0.179
Observations	2,860,572	2,860,572	2,860,503	2,842,344

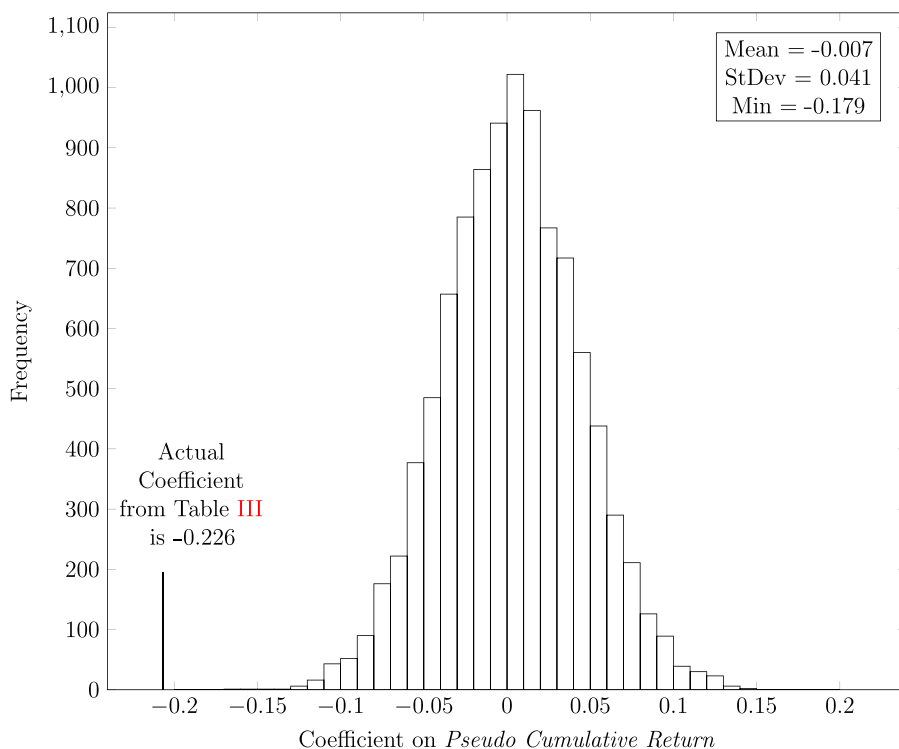


Figure 4. Coefficient on *Pseudo Cumulative Return* in placebo samples. The figure provides a histogram of *Pseudo Cumulative Return* coefficients from 10,000 iterations of the model in Table III, column (3). For each iteration, each ZIP code is randomly assigned the returns of another out-of-state ZIP code (creating new counterfactual values for *Pseudo Cumulative Return*). The model is reestimated using the counterfactual *Pseudo Cumulative Return* values. All other advisor characteristics remain the same. (Color figure can be viewed at wileyonlinelibrary.com)

from differences in purchase prices due to the time of purchase. The inclusion of residency length $\times$ year fixed effects removes the average across-ZIP code variation arising from time of purchase.

This specification is quite conservative and likely removes some of the variation of interest, but it eliminates potential confounding effects. For example, suppose that customers' wealth affects their vigilance, and that detection of misconduct is negatively related to housing returns. This could create a spurious relation between housing returns and misconduct, but it would be removed by the inclusion of the ZIP code-year fixed effects.

The results in column (1) show that, even after including ZIP code-year fixed effects, the relation between misconduct and cumulative housing returns remains negative and significant. Furthermore, the magnitude of the coefficient is similar to that in the baseline specification.

B. High-Dimensional Fixed Effects

To address the concern that advisors who work in a specific area may have some commonality due to within-firm variation such as managerial discipline (e.g., Egan, Matvos, and Seru (2020)), we include specifications to control for branch-level commonality. We classify all advisors who are employed by the same firm and work in the same county as working in a branch. This definition of branch follows other recent papers using BrokerCheck data (e.g., Charoenwong, Kwan, and Umar (2019), Egan, Matvos, and Seru (2020)).²⁹

Column (2) of Table IV, Panel A includes branch-year fixed effects, which subsume the firm-year fixed effects in the baseline specification. The branch-year fixed effects remove any variation common to advisors who work in the same county for the same firm during the year, including local housing price shocks, the economic circumstances of the firm's local customer base, local monitoring and oversight, and the firm's product offerings. Even with these more stringent fixed effects, the results show a significant relation between housing returns and misconduct. Column (3) of Table IV, Panel A further refines the specification by including branch-year-city fixed effects. This specification effectively limits the comparison to advisors who work for the same firm and live in the same city during the year. As we show in Figure 2, there can be substantial variation in housing price returns even within the same city. We take advantage of this variation to show that even for advisors who face similar firm and geographic environments, a significant negative relation exists between *Cumulative Return* and misconduct.

The most stringent specification, in column (4) of Table IV, Panel A, includes branch-year-ZIP code fixed effects, which subsume both the firm-year and ZIP code fixed effects. This specification effectively limits the comparison to advisors who work in the same county for the same firm and live in the same ZIP code during the year. These are very restrictive fixed effects that remove many sources of potentially confounding variation. The cost of including these fixed effects, however, is that we likely remove variation of interest and we lose more than 60% of the observations due to insufficient variation within the fixed effect unit. The results show that, once again, there is a significant negative relation between *Cumulative Return* and misconduct. Advisors' housing returns affect misconduct even relative to their local coworkers.

Beyond economic conditions, another potential concern is that advisors sort into branches based on demographic characteristics. In Internet Appendix Table IA.XI, we include firm-year-city-demographic characteristic (age, tenure, or urban-rural classification) fixed effects. Across all specifications, the coefficient is of similar magnitude and is statistically significant.

²⁹ This definition differs from that of Dimmock, Gerken, and Graham (2018), who use exact business address information that is not provided by all states. A firm may have different unique business addresses within the same county, and the data we use in our paper do not differentiate these unique offices.

C. Out-of-State Customers

The previous fixed effects specifications remove variation common to advisors living in the same ZIP code or working at the same firm during a year—such as shared variation in the advisors' customer base. In this section, we take a different approach and address the issue of advisor-customer commonality by using an alternative dependent variable. Although many customer-advisor matches are between geographically proximate individuals, this is not always the case. In the data, we observe the state of residence for customers who file complaints and find that 15.4% of customer complaints are filed by out-of-state customers.³⁰ In column (1) of Table IV, Panel B, we limit the dependent variable to customer complaints filed by out-of-state customers. In column (2), we further limit the dependent variable to exclude complaints filed by customers living in bordering states (e.g., for advisors living in Florida we exclude out-of-state complaints filed by customers living in Alabama and Georgia).

In both columns, the results show that the relation between the advisor's *Cumulative Return* and misconduct remains significant even for these restricted dependent variables. The coefficient estimates on *Cumulative Return* are lower than in the prior specifications because out-of-state misconduct is a subset of all misconduct, but the implied economic magnitudes as a percentage change in the base rate of the dependent variable are similar to that in column (3) of Table III. These tests break the link between the local real estate return of the advisor and the return of the (geographically distant) customer, and therefore suggest that it is not the customers' real estate returns that drive our results.

D. Regulatory Actions and Employment Separation after Allegations

In this section, we present another test to address the potential concern that, even after the inclusion of the fixed effects, advisors' *Cumulative Returns* are correlated with their customers' housing returns, or more generally, with time-series variation in customers' propensity to file complaints. Specifically, in columns (3) and (4) of Table IV, Panel B, the dependent variable is an indicator for whether the advisor discloses a regulatory action or an employment separation after allegations. We exclude all advisor-year observations that include a customer complaint to ensure the regulatory actions and employment separations are not responses to customer complaints (e.g., if a firm terminated an advisor because of a customer complaint).³¹

Government regulation of financial advisors is conducted by state governments. We include state-year fixed effects to remove the state regulator's

³⁰ As multiple customers can file a complaint in a given year, 27.3% of advisor-years with *Misconduct* include at least one out-of-state complaint.

³¹ We hand checked a subsample of regulatory actions and employment separations that occur without customer complaints. Examples of the stated reasons include "incorrect reporting of information into company system," "holding nonapproved seminars," "sale of nonapproved products," and "use of nonapproved marketing materials."

overall propensity to take actions in a given year.³² We continue to include firm-year fixed effects to remove each firm's propensity to terminate its employees during a given year. These fixed effects remove the most obvious sources of confounding variation, as there is no obvious reason to expect that a state regulator or firm would disproportionately target advisors based on cumulative housing returns.

The results in columns (3) and (4) show that even when misconduct is limited to exclude customer complaints, a significant and negative relation exists between *Cumulative Return* and misconduct. Overall, the results in both panels of Table IV support the view that housing returns affect the propensity of financial advisors to commit misconduct.

V. Robustness Tests: Alternative Dependent Variables

A. Active versus Passive Misconduct

Throughout the paper, we have interpreted misconduct as an *active* choice, whereby advisors deliberately exploit their clients for financial gain. In this section, we consider an alternative possibility—that real estate shocks result in “passive” misconduct through inattention or negligence. Passive misconduct could increase following real estate shocks if the advisor becomes distracted and less effective at work due to financial pressure (e.g., Maturana and Nickerson (2020) show that students' standardized test scores suffer when their teacher suffers financial distress). Note that active and passive misconduct are not mutually exclusive, and both could occur for the same individual.

We separate misconduct into active and passive categories by parsing the text fields in the advisors' disclosure statements and classifying misconduct based on key words. The variable *Active Misconduct* includes misrepresentation, unauthorized trading, fee or commission-related misconduct, churning, and fraud. These are acts of commission (intentional actions) that enrich the advisor if undetected. The variable *Passive Misconduct* includes negligence and omission of key facts. These are acts of omission, consistent with carelessness or inattention. We do not classify some categories of misconduct, such as unsuitability and violations of fiduciary duty. Of advisor-years with misconduct events, 53.6% are classified as active, 17.0% are classified as passive, and the remainder cannot be unambiguously classified.

In column (1) of Table V, the dependent variable includes only *Active Misconduct*. The coefficient on *Cumulative Return* is significant and negative. This result indicates that the negative relation between housing returns and misconduct holds for active misconduct—deliberate actions taken to exploit customers for financial gain—and thus our main results are not driven entirely by inattention or negligence. The result in column (2) indicates that there is also evidence of passive misconduct. The coefficient on active misconduct

³² FINRA is responsible for some regulation at the national level, but national-level effects will be subsumed by the state-year fixed effects.

Table V
Alternative Measures of Misconduct

This table reports estimates from regressions of alternative measures of misconduct on an advisor's cumulative house price return since purchase. The unit of observation is the advisor-year. The dependent variables are listed above each column and are indicators for different types of advisor misconduct. *Active* and *Passive* are misconduct measures separated out by type. We define *Active* misconduct as customer disputes related to misrepresentation, unauthorized activity, fees/commissions, churning/excessive trading, and fraud. We define *Passive* misconduct as customer disputes related to negligence and omission of key facts. *Mutual Fund* indicates whether the misconduct was related to mutual funds. $\geq 100k$ indicates whether alleged damages exceed \$100,000. *EMS* is the misconduct measure of Egan, Matvos, and Seru (2019). In each column, we multiply the dependent variable by 100. *Cumulative Return* is the aggregate cumulative price change divided by the purchase price. $\log(\text{Industry Experience})$ is the logarithm of the number of years an advisor has worked in the industry. Indicators for fixed effects are reported beneath the estimates. Standard errors, clustered by advisor and ZIP code, are reported in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% level, respectively.

	Active (1)	Passive (2)	Mutual Fund (3)	$\geq 100k$ (4)	EMS (5)
Cumulative Return	−0.0925*** (0.0238)	−0.0437*** (0.0141)	−0.0827*** (0.0256)	−0.0815*** (0.0196)	−0.2836*** (0.0381)
$\log(\text{Industry Experience})$	0.2285*** (0.0160)	0.0750*** (0.0088)	0.4745*** (0.0170)	0.1705*** (0.0130)	0.5307*** (0.0269)
Advisor FE	Yes	Yes	Yes	Yes	Yes
Residency Length×Year FE	Yes	Yes	Yes	Yes	Yes
Firm×Year FE	Yes	Yes	Yes	Yes	Yes
ZIP FE	Yes	Yes	Yes	Yes	Yes
R^2	0.140	0.131	0.126	0.135	0.163
Observations	2,860,572	2,860,572	2,860,572	2,860,572	2,860,572

is marginally significantly larger than the coefficient on passive misconduct (p -value = 0.079).

B. Alternative Definitions of Misconduct

In this subsection, we present results of several robustness tests that are similar to the baseline specification but use alternative definitions of misconduct. In column (3) of Table V, the dependent variable is an indicator variable denoting advisors who commit misconduct involving mutual funds. Mutual funds are a relatively straightforward financial product, and are typically regulated and distributed at the national level. In contrast, other financial products may vary across states due to regulator differences (e.g., annuities and some insurance products) or may be distributed primarily in a limited geographic area (e.g., local microcap stocks). Limiting the dependent variable to include only mutual funds reduces the possibility that some type of assortative matching of local products to local customers could confound our analysis. Consistent with the baseline specification, column (3) finds a significant negative coefficient on *Cumulative Return*.

In column (4) of Table V, the dependent variable is an indicator variable denoting advisors who commit misconduct for which the damages exceed \$100,000. We include this specification to ensure that the relation between housing returns and misconduct holds for the most egregious cases. The results show that, even under this restrictive definition of misconduct, there is a significant negative relation between the advisor's cumulative real estate returns and large cases of misconduct.

In column (5) of Table V, we define misconduct following Egan, Matvos, and Seru (2019), who use a broader measure of misconduct that, in addition to customer complaints, includes regulatory actions, terminations by an employing firm, and criminal and civil disclosures. The results are similar to those in the baseline specification. Overall, the results in Table V show that advisors' real estate returns affect misconduct of all types.

VI. Cross-Sectional Variation in Termination Risk

In this section, we test how cross-sectional variation in termination risk affects the relation between real estate shocks and misconduct. Prior studies show that there is large variation in the likelihood that an advisor is terminated after committing misconduct (e.g., Egan, Matvos, and Seru (2020, 2019)). Higher termination risk implies a lower expected return to committing misconduct. Thus, all else equal, we expect the relation between *Cumulative Return* and misconduct to be weaker for advisors with higher termination risk.

Egan, Matvos, and Seru (2019) find large across-firm variation in tolerance for misconduct. We measure each advisor's termination risk based on the fraction of the other advisors at the firm who have a history of prior misconduct.³³ If this fraction is above the sample average, we set the indicator variable *High Firm-Year* equal to one.³⁴ The results, reported in column (1) of Table VI, show that the coefficient on the interaction term *Cumulative Return* $\times$ *High Firm-Year* is significant and negative. Even after controlling for firm-year fixed effects, the effect of a real estate shock on misconduct is greater when the career risk associated with committing misconduct is smaller.

The specification in column (2) of Table VI is conceptually similar to that in column (1), but here we consider tolerance for misconduct at a finer geographic level. For this test, we set the indicator variable *High Branch-Year* to one if an above-average proportion of the advisor's fellow employees who work in the

³³ Prior misconduct is any disclosed substantiated customer complaint occurring before the current year.

³⁴ We do not include the advisor's own history of misconduct when constructing this variable because of biases that occur in fixed effects regressions in which an independent variable is a function of lagged values of the dependent variable. See Nickell (1981) for further discussion. Because the advisor's own history of misconduct is not included in this variable, its direct effect is not fully subsumed by the firm-year fixed effects and thus it is included in the regression. However, due to its high correlation with the firm-year fixed effects, the coefficient on the direct effect is unreliable due to multicollinearity. Notwithstanding, the issue of multicollinearity does not affect the coefficient on the interaction term, which is the coefficient of interest.

Table VI
Cross-Sectional Variation in Termination Risk

This table reports estimates from regressions of measures of misconduct on an advisor's cumulative house price return since purchase. The unit of observation is the advisor-year. The dependent variable is an indicator for advisor misconduct (multiplied by 100). *Cumulative Return* is the aggregate cumulative price change divided by the purchase price. $\log(\text{Industry Experience})$ is the logarithm of the number of years an advisor has worked in the industry. *High Firm-Year* is an indicator set to one if the firm-year has an above average misconduct rate (excluding the advisor's own misconduct). *High Branch-Year* is an indicator set to one if the branch-year has an above average misconduct rate (excluding the advisor's own misconduct). *Branch* is defined as a firm-county. *<10 Employees* is an indicator set to one if the firm employs fewer than 10 advisors. *Female* is an indicator for gender. *Prior Misconduct* is an indicator set to one if the advisor has misconduct in previous years. Column (5) includes the following control variables: *Series 65*, *Series 66*, *Series 24*, *Series 7*, *Series 6*, *Age*, and *Female*. *Series 65* equals one if the advisor obtained a license to act as an investment advisor, *Series 66* equals one if the advisor obtained a license to act as an investment advisor or securities agent, *Series 24* equals one if the advisor obtained a license to act as a supervisor, *Series 7* equals one if the advisor obtained a license to sell securities, *Series 6* equals one if the advisor obtained a license to sell mutual funds, variable annuities, and insurance premiums, and *Age* is the advisor's age in years. Indicators for fixed effects are reported beneath the estimates. Standard errors, clustered by advisor and ZIP code, are reported in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% level, respectively.

	(1)	(2)	(3)	(4)	(5)
Cumulative Return	-0.1807*** (0.0336)	-0.1532*** (0.0340)	-0.2250*** (0.0335)	-0.3051*** (0.0385)	-0.0847*** (0.0225)
High Firm-Year	5.1991*** (0.3923)				
C.Ret. × High Firm-Year	-0.0864* (0.0466)				
High Branch-Year		0.0664*** (0.0239)			
C.Ret. × High Branch-Year		-0.1535*** (0.0446)			
C.Ret. × <10 Employees			-0.3508** (0.1768)		
C.Ret. × Female				0.3049*** (0.0424)	
Prior Misconduct					2.1525*** (0.0649)
C.Ret. × Prior Misconduct					-0.3510*** (0.1266)
$\log(\text{Industry Experience})$	0.4849*** (0.0225)	0.4859*** (0.0225)	0.4869*** (0.0225)	0.4874*** (0.0226)	0.2378*** (0.0079)
Advisor FE	Yes	Yes	Yes	Yes	No
Advisor Controls	No	No	No	No	Yes
Residency Length × Year FE	Yes	Yes	Yes	Yes	Yes
Firm × Year FE	Yes	Yes	Yes	Yes	Yes
ZIP FE	Yes	Yes	Yes	Yes	Yes
R^2	0.146	0.146	0.146	0.146	0.032
Observations	2,860,572	2,860,572	2,860,572	2,843,273	2,882,302

same county for the same firm have a history of past misconduct. This variable allows for the possibility that a firm's tolerance for misconduct in a given year could vary across branches due to local oversight, state regulators, or other reasons. The coefficient on the interaction term is significant and negative. Thus the negative relation between housing returns and misconduct is stronger for advisors who work for a branch that has a high tolerance for misconduct.

Column (3) of Table VI includes an interaction term between *Cumulative Return* and *<10 Employees*, an indicator variable equal to one for advisors working at firms with fewer than 10 employees. Firms with only a few employees likely have less formalized internal monitoring procedures and are also less likely to terminate employees following misconduct. The coefficient on the interaction term is significant and negative, indicating that the effect of real estate shocks on misconduct is stronger for advisors with lower career risk.

Column (4) of Table VI considers individual-level career risk from committing misconduct, rather than firm- or branch-level career risk. Egan, Matvos, and Seru (2020) show that, relative to men, women are more likely to be terminated for misconduct and less likely to find new employment conditional on termination. Accordingly, women face greater career risk for committing misconduct. As in Egan, Matvos, and Seru (2020), 25% of advisors in our sample are female. The coefficient on the interaction term *Cumulative Return* $\times$ *Female* is significant and positive. Indeed, for females the net effect of *Cumulative Return* is not significant. Overall, the results in the first four columns show that the relation between real estate shocks and misconduct is stronger when the career risk from committing misconduct is lower.

Column (5) of Table VI includes an interaction term between *Cumulative Return* and *Prior Misconduct*, which is an indicator variable equal to one for advisors with any prior incident of misconduct (occurring at any time in the advisor's history prior to the current year). In this specification, we do not include the advisor fixed effects so as to avoid the Nickell (1981) bias. We do, however, include advisor control variables: experience, gender, age, and licenses held. The coefficient on *Cumulative Return* is negative and significant, indicating that housing shocks affect misconduct for the full sample of advisors, including those without a record of prior misconduct. As expected, the coefficient on *Prior Misconduct* is positive and significant, indicating that past offenders are 2.15 percentage points more likely to commit misconduct. The coefficient on the interaction term is negative and significant, indicating that advisors with a history of prior misconduct are significantly more sensitive to changes in housing prices. Indeed, the magnitude of the change in misconduct for advisors with prior misconduct is more than six times larger than for advisors with no prior misconduct.

Advisors with prior misconduct may be more sensitive to real estate shocks for several reasons. First, these advisors may perceive a relatively low cost to misconduct given that they have remained employed within the industry despite prior misconduct. Second, studies in the economics of crime suggest that offenders have different preferences than the general population, in particular, offenders tend to be myopic and place little weight on deferred

punishments (Lee and McCrary (2005)). Given that, in our setting, misconduct produces immediate gains in return for uncertain delayed punishments, *Prior Misconduct* may indicate advisors with high discount rates whose misconduct decisions are particularly sensitive to changes in collateral values.

VII. Examining Potential Mechanisms

A. Borrowing Constraints and Misconduct

Lott (1990), Andreoni (1992), and Kotsogiannis and Mateos-Planas (2019) argue that misconduct can serve as a form of contingent borrowing: misconduct provides immediate funds but may require a large future payment if detected. In our setting, an advisor who wants to borrow can compare the costs of conventional borrowing to “borrowing” via misconduct and choose the cheaper option. Mian, Rao, and Sufi (2013), Aladangady (2017), and Cloyne et al. (2019) show that changes in home prices have large effects on households’ borrowing constraints. These studies further show that the sensitivity of borrowing to changes in home prices is stronger for individuals with higher borrowing constraints. Although we cannot directly observe advisors’ borrowing constraints, we create two proxies: advisors with large negative cumulative returns and advisors with high imputed LTV ratios. We expect that the relation between misconduct and housing returns will be stronger for constrained advisors.

The specification in column (1) of Table VII includes the interaction term (*Cumulative Return* $- (-20\%)$) $\times I_{Extreme}$, where $I_{Extreme}$ indicates a cumulative return below -20% . We use a cutoff of -20% to identify advisors who potentially face borrowing constraints, as this indicates a decline below the commonly recommended down payment. Furthermore, many homeowners likely have only an imprecise estimate of their home’s value based on their neighbors’ home sales, local news stories, etc.

The specification in column (2) of Table VII includes the interaction term *Cumulative Return* $\times I_{HighLTV}$, where $I_{HighLTV}$ indicates an imputed LTV ratio greater than 90. In our data, we do not observe mortgage amounts, refinancings, or loan terms, and so we impute LTV ratios as of the beginning of the year following a method similar to that in Bernstein, McQuade, and Townsend (2021). Specifically, we assume that, at purchase, advisors make a down payment of 20% and enter into a 30-year fixed rate mortgage at the average 30-year mortgage rate during that month (taken from the Federal Reserve Economic Database). We then amortize the loan to calculate the implied mortgage balance, which is divided by the beginning-of-year home price. By imputing LTV ratios using standardized assumptions, we remove the endogenous variation in LTV ratios caused by individuals’ choices (aside from the time-of-purchase decision, which is controlled for by residency length $\times$ year fixed effects). Using these imputed LTV ratios, we follow Bernstein, McQuade, and Townsend (2021) and use a cutoff of LTV ratios greater than 90 to create *High LTV*, an indicator variable equal to one for 5.62% of the advisors in our sample. Relative to the $I_{Extreme}$ variable, *High LTV* considers

Table VII
Borrowing Constraints and Employee Ownership

This table reports estimates from regressions of misconduct on an advisor's cumulative house price return since purchase. The unit of observation is the advisor-year. *Cumulative Return* is the aggregate cumulative price change divided by the purchase price. $\log(\text{Industry Experience})$ is the logarithm of the number of years an advisor has worked in the industry. I_{Extreme} is an indicator equal to one when *Cumulative Return* is less than -20% . I_{HighLTV} is an indicator equal to one when the imputed LTV ratio is greater than 90. *Employee Owner* is an indicator equal to one when the advisor has an ownership stake greater than 5%. Indicators for fixed effects are reported beneath the estimates. Standard errors, clustered by advisor and ZIP code, are reported in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% level, respectively.

	(1)	(2)	(3)
Cumulative Return	-0.2098*** (0.0341)	-0.2125*** (0.0340)	-0.2115*** (0.0363)
Cumulative Return $\times$ I_{Extreme}	-0.5766** (0.2823)		
Cumulative Return $\times$ I_{HighLTV}		-0.6686*** (0.2550)	
High LTV		-0.1472** (0.0668)	
Cumulative Return $\times$ Employee Owner			-0.9407** (0.3951)
$\log(\text{Industry Experience})$	0.4868*** (0.0225)	0.4869*** (0.0225)	0.4604*** (0.0237)
Advisor FE	Yes	Yes	Yes
Residency Length $\times$ Year FE	Yes	Yes	Yes
Firm $\times$ Year FE	Yes	Yes	Yes
ZIP FE	Yes	Yes	Yes
R^2	0.146	0.146	0.161
Observations	2,860,572	2,860,572	2,521,316

the possibility of mortgage paydown on borrowing constraints, albeit at the cost of introducing measurement error through the imputation assumptions.

Consistent with the misconduct-as-borrowing argument, the coefficients on the interaction terms in columns (1) and (2) are negative and significant. The relation between housing returns and misconduct is stronger for advisors whose ability to borrow against home equity is likely more constrained, as measured by either large negative *Cumulative Returns* or high imputed LTV ratios. Furthermore, the economic magnitudes of the results are large. For example, the results in column (2) suggest that the economic magnitude of a one standard deviation shock to *Cumulative Return*, considering both the interaction term and the direct effect, is about twice as large for advisors with an LTV ratio greater than 90.

In both columns, the direct effect of *Cumulative Return* remains negative and significant even with the inclusion of the interaction term. Indeed, as [Internet Appendix Table IA.XII](#) shows, even high positive values of

Cumulative Return have a significant relation with misconduct.³⁵ This pattern of results is consistent with misconduct as risky borrowing. During our sample period, borrowing against home equity was common throughout the range of cumulative returns (see Mian and Sufi (2011), Bhutta and Keys (2016)) and the marginal propensity to consume out of housing equity occurred over the full range of housing returns, albeit more strongly for constrained households. This pattern is also consistent with ethical preferences. If advisors derive utility from ethical behavior and ethical behavior is a normal good, then advisors' willingness to commit misconduct will be inversely related to wealth (see Block and Heineke (1975), Gordon (1989)) and this relation will operate over the entire range of housing returns, including large positive returns, albeit more strongly for low returns due to utility curvature. This pattern is inconsistent with various other mechanisms including those based on financial distress, for which the effect would occur only for very low *Cumulative Returns*.

B. Employee Ownership

There are two types of penalties for detected misconduct: labor market (e.g., Egan, Matvos, and Seru (2019)) and financial. The direct financial penalties for detected misconduct can be discharged in bankruptcy. Indeed, a FINRA study of arbitration cases in the 2012 to 2016 period found that awarded damages went unpaid in 27% of cases, frequently because of bankruptcy.³⁶ As Block and Lind (1975) show, when financial penalties can be discharged in bankruptcy, their effectiveness in deterring misconduct falls as wealth declines. Conceptually, this is similar to the classic asset substitution problem of Jensen and Meckling (1976).

In the financial advisory industry, the penalties for misconduct are paid largely by the firms.³⁷ There is, however, a set of advisors who bear the direct financial risk of misconduct penalties—employee-owners.³⁸ For employee-owners, their home values directly affect their ability to pay financial penalties (and their bargaining power when negotiating settlements). We therefore expect the relation between housing returns and misconduct to be stronger for employee-owners. We identify employee-owners using data from a cross-section of Schedules A and B of FINRA Form BD.³⁹ For this test, we drop observations for firms for which we do not have Form BD data.

³⁵ Internet Appendix Table IA.XII shows interaction results using indicators for +20% (positive instead of negative) and +90% (the 95th percentile of the distribution instead of the 5th). The effect of *Cumulative Returns* is not significantly diminished for large positive returns.

³⁶ The full study is available at https://www.finra.org/sites/default/files/finra_perspectives_on_customer_recovery.pdf.

³⁷ We obtained data detailing who pays penalties from the Florida Office of Financial Regulation as of March 16, 2011. In 76% of cases, the firm pays the entire amount. In only 6% of cases does the advisor pay the entire amount. On average, the firm pays 87.6% of the costs.

³⁸ Note that limited liability protections generally do not apply to cases of misconduct (e.g., Matheson (2010)).

³⁹ We use a cross section of Form BD filings as of August 10, 2010. Schedule A provides information on direct owners and executive officers. Schedule B provides information on indirect

The specification in column (3) of Table VII follows our baseline specification, but includes an interaction between employee-owner and *Cumulative Return*. The coefficient on the interaction term is negative and significant: the relation between housing returns and misconduct is stronger for employee-owners. This result is consistent with the effect of personal wealth exposure for professional misconduct. Housing returns have a larger effect on misconduct for advisors whose housing is directly at stake in the event of financial penalties for misconduct. However, only a small fraction of advisors are employee-owners and the coefficient on *Cumulative Return* remains significant and little changed by the inclusion of the interaction term. Thus, while this test provides some evidence that limited liability issues affect misconduct in our sample, this is not the dominant mechanism driving our results.

C. Housing Returns and Financial Advisor Wealth

C.1. The Magnitude of Housing Shocks for Financial Advisors

In this subsection, we examine the financial importance of housing wealth shocks for the financial advisors in our sample using simple back-of-the-envelope calculations. During the sample period, the standard deviation of the annual percentage change in home prices was 8.7%. Earlier, in Section VII.A, we find a median imputed LTV ratio of 71.6%. This implies that the delevered fluctuation in home equity implied by an 8.7% housing price change is 30.5%. Thus, relative to housing equity, the price fluctuations in the sample are large.

Following Bernstein, McQuade, and Townsend (2021) and Pool et al. (2019), we further consider the size of the housing fluctuation relative to imputed total wealth. Our method largely follows from Bernstein et al. First, we use data from the Bureau of Labor Statistics⁴⁰ to find the average annual earnings for individuals in the investment advice industry. In July 2007, average annual earnings totaled \$78,469. Second, we use data from the 2007 Survey of Consumer Finances (SCF). We find that the median fraction of net wealth held in housing for people in the same income decile as financial advisors is 54.3%.

Combining the standard deviation of annual home price changes, imputed LTV ratio, and median fraction of net wealth in housing implies that a one standard deviation change in annual housing prices is associated with a change in net wealth of 16.6% ($54.3\% \times 8.7\% / (1 - 71.6\%)$). Alternatively, the size of housing fluctuations implies that a one standard deviation change in the annual housing return produces a wealth effect approximately equivalent to 3.6 months average salary. Overall, the back-of-the-envelope calculations suggest that housing price fluctuations are economically meaningful for the financial advisors in our sample.

owners. See Dimmock, Gerken, and Marietta-Westberg (2015) for further discussion on classifying ownership from such schedules. We classify all individuals with ownership positions equal to or exceeding 5% of the firm as an owner. We use this cross section to assign ownership throughout the entire panel.

⁴⁰ Data are available at <https://www.bls.gov/data>.

Table VIII
Predicting Financial Distress

This table tests whether our cumulative return measure reliably predicts negative financial outcomes for advisors. The unit of observation is the advisor-year. In column (1), the dependent variable, *Bankruptcy_{t+1}*, is an indicator for whether the advisor reports a bankruptcy or other compromise with creditors in the following year. In column (2), the dependent variable, *Underwater Sale_{t+1}*, is an indicator for whether the advisor reports an underwater home sale (for which the advisor does not make the lender whole) in the following year. In both columns, the dependent variable is multiplied by 100. *Cumulative Return* is the aggregate cumulative price change divided by the purchase price. *log(Industry Experience)* is the logarithm of the number of years an advisor has worked in the industry. Indicators for fixed effects are reported beneath the estimates. Standard errors, clustered by advisor and ZIP code, are reported in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% level, respectively.

	Bankruptcy _{t+1} (1)	Underwater Sale _{t+1} (2)
Cumulative Return	-0.2616*** (0.0447)	-0.3324*** (0.0301)
log(Industry Experience)	0.0206 (0.0339)	0.0796*** (0.0172)
Advisor FE	Yes	Yes
Residency Length×Year FE	Yes	Yes
Firm×Year FE	Yes	Yes
ZIP FE	Yes	Yes
R ²	0.222	0.211
Observations	1,787,370	1,787,370

C.2. Cumulative Housing Returns and Advisor Financial Distress

Throughout the paper, we use imputed house price returns to measure shocks to financial advisors’ borrowing constraints. As a validation check of our *Cumulative Return* variable, we consider an extreme example of borrowing constraints, namely, that of financial distress. FINRA requires advisors to disclose any bankruptcy filing or other “compromise with creditors” in which “a creditor agrees to accept less than the full amount owed”⁴¹ for a 10-year period.⁴² We use these disclosures to create two measures of financial distress.

In column (1) of Table VIII, the dependent variable is an indicator variable denoting advisors who file for bankruptcy in the next year. Aside from this change to the dependent variable, the specification is identical to the baseline specification. The results show that advisors with worse cumulative housing returns are significantly more likely to declare bankruptcy. Furthermore, the implied economic magnitude of the estimate is large; the coefficient estimate implies that a one standard deviation change in *Cumulative Return* is

⁴¹ See www.finra.com/finra/wp-content/uploads/2015/04/Finra-U4-U5-QnA.pdf for a more detailed definition of the reporting requirements.

⁴² Because advisors can cease disclosing events after 10 years, we supplement the BrokerCheck data using financial distress disclosures reported in FOIA requests ca. 2011.

associated with a 0.11 percentage point change in the probability of bankruptcy (a 36.4% change relative to the baseline rate).

In column (2) of Table VIII, the dependent variable is an indicator variable denoting advisors who disclose a “compromise with creditors” related to an underwater sale (a house sale in which the proceeds are less than the debts secured by the property and the advisor does not pay the lender the difference) in the next year. This provides a direct measure of financial distress related to housing returns. The results show that advisors with worse cumulative house returns are significantly more likely to disclose underwater home sales. Furthermore, the implied magnitude is large: the coefficient estimate implies that a one standard deviation change in *Cumulative Return* is associated with a 0.14 percentage point change in underwater sales (a 130.4% change relative to the baseline rate). Overall, the results in Table VIII provide evidence that our imputed measure of housing price returns captures meaningful wealth shocks for financial advisors.

VIII. Conclusion

In this paper, we test whether personal wealth shocks affect the propensity of employees to engage in professional misconduct. We measure financial shocks using financial advisors’ housing price returns, and we measure misconduct using validated customer complaints disclosed in mandatory regulatory filings. We find a negative relation between housing returns and misconduct, and many of these misconduct events are willful actions such as churning and unauthorized trading. We also show that advisors’ housing returns explain misconduct against out-of-state customers, breaking the link between customer and advisor housing shocks. Furthermore, the results are stronger for advisors with lower career risk from committing misconduct, and for advisors with greater borrowing constraints. Our finding that changes in personal circumstances change the propensity to commit professional misconduct implies a need for regulators to require ongoing disclosure of personal information.

Our findings are consistent with misconduct as a form of risky borrowing. Many types of financial misconduct entail both a risky and an intertemporal choice, in that the rewards of misconduct occur before the potential penalties. In our setting, advisors who wish to borrow can choose between conventional loans, such as home equity extraction, and “borrowing” via misconduct, as in the model of Andreoni (1992), with changes in collateral values affecting this choice. Demonstrating the empirical relevance of misconduct as borrowing has implications for policy design. First, it implies that the speed of detection and resolution is important, as faster resolution effectively increases the cost of borrowing via misconduct. Second, it implies a role for prejudgment interest (interest added to assessed damages for the period from the commission of misconduct until judgment). Making prejudgment interest mandatory and setting a high interest rate would provide a disincentive for “borrowing” via misconduct.

Our finding that misconduct is negatively related to wealth suggests important welfare implications. Although our empirical tests difference out aggregate time-series effects, to the extent that wealth effects on misconduct generalize to aggregate wealth shocks, there are two implications. First, the willingness to commit misconduct will rise in states of the world in which marginal utility is highest, making the direct losses from misconduct particularly painful. Second, misconduct erodes trust and causes people to exit markets (see Giannetti and Wang (2016), Gurun, Stoffman, and Yonker (2018)) precisely when expected returns are the highest.

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Supporting Information

Additional Supporting Information may be found in the online version of this article at the publisher's website:

Appendix S1: Internet Appendix.
Replication Code.